

# Chapter 4 Test model

- **70 Multiple Choice Questions (MCQs)**
  - **30 True/False Questions**
  - **Difficulty: 70% Hard, 30% Very Hard**
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## Multiple Choice Questions (1–70)

1. In EER modeling, a subclass that inherits from more than one superclass is known as a:
  - a. Disjoint subclass
  - b. Shared subclass
  - c. Predicate-defined subclass
  - d. Union type

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2. The relationship "SECRETARY IS-A EMPLOYEE" exemplifies which of the following?
  - a. Generalization
  - b. Aggregation
  - c. IS-A relationship
  - d. Categorization

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3. Which of the following is NOT an EER extension beyond basic ER modeling?
  - a. Attribute inheritance
  - b. Weak entities
  - c. Specialization
  - d. Categories (UNION types)

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4. In an attribute-defined specialization, the attribute used to determine subclass membership is called the:
  - a. Local attribute
  - b. Inherited attribute
  - c. Defining attribute
  - d. Composite attribute

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5. A specialization where an entity can belong to more than one subclass is termed:
- a. Total
  - b. Disjoint
  - c. Overlapping
  - d. Partial
- 

6. In EER diagrams, a double line connecting a superclass to its subclasses indicates:
- a. Disjoint constraint
  - b. Overlapping constraint
  - c. Total completeness
  - d. Partial completeness
- 

7. Which symbol is used in EER diagrams to denote a disjoint constraint?
- a. o
  - b. d
  - c.  $\cap$
  - d.  $\cup$
- 

8. The process of defining subclasses of a superclass based on distinguishing characteristics is called:
- a. Generalization
  - b. Abstraction
  - c. Specialization
  - d. Categorization
- 

9. A category (UNION type) differs from a shared subclass in that a category member must exist in:
- a. All of its superclasses
  - b. None of its superclasses
  - c. At least one of its superclasses
  - d. Exactly two of its superclasses
- 

10. Inheritance in EER modeling allows a subclass to inherit:
- a. Only attributes from the superclass
  - b. Only relationships from the superclass

- c. Both attributes and relationships from the superclass
  - d. Neither attributes nor relationships
- 

11. Which of the following best describes generalization?

- a. Top-down refinement
  - b. Bottom-up synthesis
  - c. Horizontal partitioning
  - d. Vertical fragmentation
- 

12. If a specialization uses the same defining attribute for all its subclasses, it is called:

- a. User-defined
  - b. Predicate-defined
  - c. Attribute-defined
  - d. Condition-independent
- 

13. In a specialization lattice, a subclass may have:

- a. Only one direct superclass
  - b. Multiple direct superclasses
  - c. No superclasses
  - d. Only leaf superclasses
- 

14. The term “multiple inheritance” in EER modeling refers to:

- a. A superclass inheriting from multiple subclasses
  - b. A subclass inheriting from multiple superclasses
  - c. Overlapping specializations with shared attributes
  - d. Recursive generalization
- 

15. Which of the following is a valid reason for using a category instead of a shared subclass?

- a. When the superclasses represent disjoint entity types
  - b. When the subclass must be a member of all superclasses
  - c. When the subclass represents a union of heterogeneous superclasses
  - d. When inheritance must be single-parent only
-

16. In the EER model, local attributes of a subclass are:

- a. Inherited from the superclass
  - b. Shared with other subclasses
  - c. Specific to that subclass
  - d. Always composite
- 

17. Consider an EMPLOYEE superclass with subclasses ENGINEER, MANAGER, and SALARIED\_EMPLOYEE. An employee who is both a manager and an engineer belongs to:

- a. Only MANAGER
  - b. Only ENGINEER
  - c. Both MANAGER and ENGINEER
  - d. None, due to disjoint constraint
- 

18. Which constraint ensures that every entity in the superclass is a member of at least one subclass?

- a. Disjointness
  - b. Overlapping
  - c. Total completeness
  - d. Partial inheritance
- 

19. In EER diagrams, user-defined subclasses are typically indicated by:

- a. Absence of a predicate label
  - b. A "u" symbol
  - c. Dashed lines
  - d. No standard notation—implied by context
- 

20. A specialization hierarchy enforces:

- a. Multiple inheritance
  - b. Single inheritance
  - c. Total completeness only
  - d. Overlapping membership
- 

21. Which of the following is true about shared subclasses?

- a. They represent the union of their superclasses

- b. They must be members of all their superclasses
  - c. They cannot have local attributes
  - d. They violate the IS-A principle
- 

22. The EER model supports which type of inheritance?

- a. Only single inheritance
  - b. Only multiple inheritance
  - c. Both single and multiple inheritance
  - d. Neither—inheritance is not supported
- 

23. In a UNIVERSITY EER schema, if GRADUATE\_STUDENT is a subclass of both STUDENT and EMPLOYEE, this is an example of:

- a. Attribute-defined specialization
  - b. Disjoint hierarchy
  - c. Shared subclass (multiple inheritance)
  - d. Category (UNION type)
- 

24. Which of the following is NOT a type of specialization constraint?

- a. Disjoint, total
  - b. Overlapping, partial
  - c. Predicate, recursive
  - d. Disjoint, partial
- 

25. What does the "o" symbol denote in an EER diagram?

- a. Total completeness
  - b. Overlapping subclasses
  - c. Optional participation
  - d. Object-oriented class
- 

26. Generalization is typically:

- a. Partial and disjoint
- b. Total and overlapping
- c. Total because the superclass is derived from subclasses
- d. Always user-defined

- 
27. In EER modeling, a subclass that does not include all superclass entities demonstrates:
- a. Total completeness
  - b. Partial completeness
  - c. Disjointness
  - d. Attribute inheritance
- 

28. Which of the following best describes a predicate-defined subclass?
- a. Membership is assigned by the DBA per entity
  - b. Membership is determined by a condition on a superclass attribute
  - c. Membership requires intersection of multiple superclasses
  - d. Membership is based on external ontologies
- 

29. In the vehicle registration example, OWNER is a category because:
- a. It inherits from VEHICLE
  - b. It is a shared subclass of PERSON, BANK, and COMPANY
  - c. It represents the union of PERSON, BANK, and COMPANY
  - d. It is a superclass of REGISTERED\_VEHICLE
- 

30. Local relationships in a subclass are:
- a. Inherited from the superclass
  - b. Available to all subclasses
  - c. Specific to that subclass only
  - d. Always total participation
- 

31. Which statement is correct regarding inheritance in EER?
- a. Subclasses inherit only key attributes
  - b. Superclasses inherit local attributes from subclasses
  - c. Subclasses inherit all attributes and relationships of the superclass
  - d. Inheritance applies only to relationships, not attributes
- 

32. A specialization with defining attribute JobType and subclasses {SECRETARY, ENGINEER, TECHNICIAN} is:
- a. User-defined and disjoint

- b. Attribute-defined and possibly overlapping
  - c. Predicate-defined and total
  - d. Category-based
- 

33. In a lattice structure, a subclass can:
- a. Have only one ancestor
  - b. Participate in multiple IS-A hierarchies
  - c. Not inherit from more than one level
  - d. Only be leaf-level
- 

34. The primary purpose of a category (UNION type) is to:
- a. Enforce disjointness
  - b. Model a subclass of heterogeneous superclasses
  - c. Replace generalization
  - d. Simplify attribute inheritance
- 

35. Which of the following is a key difference between generalization and specialization?
- a. Generalization is top-down; specialization is bottom-up
  - b. Specialization creates superclasses; generalization creates subclasses
  - c. Generalization is bottom-up; specialization is top-down
  - d. They are identical in EER modeling
- 

36. In EER diagrams, how is an attribute-defined specialization typically displayed?
- a. With the defining attribute name near the superclass-subclass line
  - b. With a double diamond
  - c. With a "U" symbol
  - d. With dashed ellipses
- 

37. If an EMPLOYEE can be both HOURLY\_EMPLOYEE and SALARIED\_EMPLOYEE, the specialization must be:
- a. Disjoint and total
  - b. Overlapping and partial
  - c. Disjoint and partial
  - d. Overlapping and total

- 
38. Which of the following is true about the IS-A relationship?
- a. It implies the subclass is a different real-world entity
  - b. It allows an entity to exist only in the subclass
  - c. The subclass and superclass represent the same real-world entity in different roles
  - d. It prohibits attribute inheritance
- 

39. In a shared subclass like ENGINEERING\_MANAGER (subclass of both MANAGER and ENGINEER), the entity must:
- a. Belong to either MANAGER or ENGINEER
  - b. Belong to both MANAGER and ENGINEER
  - c. Belong to neither
  - d. Be a category
- 

40. What is the inheritance path in a specialization lattice?
- a. Only from immediate superclass
  - b. From all predecessor superclasses
  - c. Only from the root superclass
  - d. No inheritance in lattices
- 

41. Which of the following constraints is most likely applied to a generalization?
- a. Partial and disjoint
  - b. Overlapping and partial
  - c. Total and disjoint
  - d. Total (due to derivation from subclasses)
- 

42. A subclass with no local attributes or relationships is still meaningful if:
- a. It enables constraint modeling (e.g., disjointness)
  - b. It has a composite key
  - c. It participates in a category
  - d. It is user-defined
- 

43. In EER modeling, the term “class” can refer to:
- a. Only superclasses

- b. Only subclasses
  - c. Entity types, superclasses, or subclasses
  - d. Only categories
- 

44. Which of the following is an example of a user-defined subclass?
- a. EMPLOYEE with JobType = 'Engineer'
  - b. STUDENT under Age > 25
  - c. A DBA manually assigning an employee to a "VIP" subclass
  - d. CAR and TRUCK generalized to VEHICLE
- 

45. The completeness constraint in specialization controls:
- a. Whether subclasses overlap
  - b. Whether superclass entities must belong to a subclass
  - c. Whether inheritance is single or multiple
  - d. Whether attributes are atomic
- 

46. In an EER diagram, a single line from superclass to subclasses indicates:
- a. Total completeness
  - b. Partial completeness
  - c. Disjointness
  - d. Overlapping
- 

47. Which of the following scenarios requires a category rather than a shared subclass?
- a. A GRADUATE\_STUDENT who is both STUDENT and EMPLOYEE
  - b. An OWNER who can be a PERSON, BANK, or COMPANY
  - c. A TECHNICIAN who is also SALARIED\_EMPLOYEE
  - d. A CAR that is a VEHICLE
- 

48. Attribute inheritance implies that:
- a. Subclass attributes override superclass attributes
  - b. Every subclass entity has values for all inherited attributes
  - c. Superclasses gain attributes from subclasses
  - d. Only key attributes are inherited
-

49. In a disjoint, total specialization:

- a. Every superclass entity belongs to exactly one subclass
  - b. Entities may belong to multiple subclasses
  - c. Some entities may belong to no subclass
  - d. Membership is user-assigned
- 

50. The EER model was developed primarily to:

- a. Replace relational modeling
  - b. Support object-oriented databases
  - c. Enhance ER modeling with subclassing and inheritance
  - d. Simplify SQL queries
- 

51. Which of the following is a characteristic of a predicate-defined subclass?

- a. Defined by DB user per instance
  - b. Defined by a condition like Age > 30
  - c. Requires multiple inheritance
  - d. Cannot coexist with attribute-defined subclasses
- 

52. In the absence of any constraint notation in an EER diagram, the default assumption is:

- a. Disjoint and total
  - b. Overlapping and partial
  - c. Disjoint and partial
  - d. Overlapping and total
- 

53. A specialization hierarchy differs from a lattice in that a hierarchy:

- a. Allows multiple inheritance
  - b. Is always total
  - c. Enforces single inheritance (tree structure)
  - d. Cannot have shared subclasses
- 

54. The subclass MANAGER in an EMPLOYEE specialization based on role is likely:

- a. Attribute-defined with defining attribute Role
- b. User-defined
- c. Predicate-defined with condition Role = 'Manager'
- d. Any of the above, depending on design choice

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55. Which of the following is NOT inherited by a subclass?

- a. Primary key of superclass
  - b. Composite attributes of superclass
  - c. Local attributes of another subclass
  - d. Relationships of superclass
- 

56. In EER modeling, a “shared subclass” leads to:

- a. Union types
  - b. Multiple inheritance
  - c. Disjoint constraints
  - d. Partial completeness
- 

57. Categories are particularly useful when superclasses are:

- a. Homogeneous
  - b. Subclasses of the same root
  - c. Heterogeneous (e.g., PERSON, COMPANY)
  - d. Predicate-defined
- 

58. Which EER concept allows modeling an entity that plays multiple distinct roles?

- a. Weak entity
  - b. Aggregation
  - c. Subclassing with overlapping specialization
  - d. Derived attribute
- 

59. In a total specialization, if an entity is inserted into the superclass, the DBMS must:

- a. Reject the insertion
  - b. Automatically assign it to a subclass
  - c. Allow it only if it fits a subclass condition
  - d. Require explicit subclass assignment (depending on implementation)
- 

60. The notation for a category in EER diagrams typically includes:

- a. A double rectangle
- b. A circle labeled “U” or “UNION”

- c. A dashed oval
  - d. No standard symbol—contextual
- 

61. Which of the following best describes the semantic difference between intersection (shared subclass) and union (category)?
- a. Shared subclass = OR; Category = AND
  - b. Shared subclass = AND; Category = OR
  - c. Both represent AND
  - d. Both represent OR
- 

62. In an overlapping, partial specialization of EMPLOYEE into {ENGINEER, MANAGER}, an employee who is neither is:
- a. Invalid
  - b. Allowed
  - c. Automatically assigned to a default subclass
  - d. Converted to a weak entity
- 

63. Generalization is often considered a \_\_\_ process.
- a. Top-down
  - b. Deductive
  - c. Bottom-up
  - d. Recursive
- 

64. Specialization is often considered a \_\_\_ process.
- a. Inductive
  - b. Bottom-up
  - c. Top-down
  - d. Aggregative
- 

65. Which of the following is a very hard distinction in EER modeling?
- a. Between entity and attribute
  - b. Between specialization and generalization in diagrams
  - c. Between strong and weak entity
  - d. Between simple and composite attribute

- 
66. In a lattice, if class C inherits from A and B, and A and B both inherit from ROOT, then C inherits:
- a. Only from A and B
  - b. From A, B, and ROOT
  - c. Only from ROOT
  - d. Nothing due to ambiguity
- 

67. The EER model handles which of the following that basic ER cannot?
- a. Binary relationships
  - b. Cardinality constraints
  - c. Subclass-specific attributes and multiple inheritance
  - d. Weak entities
- 

68. If a specialization is both disjoint and total, the number of subclasses an entity can belong to is:
- a. Zero
  - b. One
  - c. One or more
  - d. All
- 

69. Which of the following is a potential issue with multiple inheritance in EER?
- a. Attribute name conflict from different superclasses
  - b. Violation of first normal form
  - c. Inability to represent relationships
  - d. Loss of key constraints
- 

70. In the absence of explicit constraints, EER modeling assumes subclass membership is:
- a. Disjoint and total
  - b. Overlapping and total
  - c. Overlapping and partial
  - d. Disjoint and partial
- 

## **True/False Questions (71–100)**

71. In EER modeling, an entity can exist solely as a member of a subclass without being in the superclass.
- A. True
  - B. False
- 
72. Attribute inheritance means a subclass automatically gets all attributes of its superclass.
- A. True
  - B. False
- 
73. A category is a subclass that is a subset of the intersection of its superclasses.
- A. True
  - B. False
- 
74. Generalization is the reverse process of specialization.
- A. True
  - B. False
- 
75. In a disjoint specialization, an entity may belong to more than one subclass.
- A. True
  - B. False
- 
76. The "d" symbol in EER diagrams denotes overlapping subclasses.
- A. True
  - B. False
- 
77. A shared subclass must be a member of all its superclasses.
- A. True
  - B. False
- 
78. User-defined subclasses use conditions based on attribute values.
- A. True

B. False

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79. In a total specialization, every entity in the superclass must belong to at least one subclass.

A. True

B. False

---

80. Specialization hierarchies allow multiple inheritance.

A. True

B. False

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81. A lattice structure in EER modeling supports single inheritance only.

A. True

B. False

---

82. Local attributes of a subclass are not inherited by its own subclasses.

A. True

B. False

---

83. The IS-A relationship implies identity of the real-world entity across superclass and subclass.

A. True

B. False

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84. Categories are used when superclasses are semantically similar.

A. True

B. False

---

85. In EER diagrams, a double line indicates partial completeness.

A. True

B. False

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86. Predicate-defined subclasses are a form of condition-defined specialization.

- A. True
  - B. False
- 

87. Generalization typically results in a total constraint.

- A. True
  - B. False
- 

88. A subclass can participate in relationships that its superclass does not.

- A. True
  - B. False
- 

89. All subclasses in an attribute-defined specialization must use the same defining attribute.

- A. True
  - B. False
- 

90. Inheritance in EER includes both attributes and relationships from all ancestor superclasses.

- A. True
  - B. False
- 

91. The EER model eliminates the need for foreign keys.

- A. True
  - B. False
- 

92. An overlapping specialization can also be total.

- A. True
  - B. False
-

93. A category member must exist in exactly one of its superclasses.

- A. True
  - B. False
- 

94. Specialization is a bottom-up design process.

- A. True
  - B. False
- 

95. In a UNIVERSITY EER schema, RESEARCH\_ASSISTANT as a subclass of both STUDENT and EMPLOYEE is a shared subclass.

- A. True
  - B. False
- 

96. The defining attribute in an attribute-defined specialization must be single-valued.

- A. True
  - B. False
- 

97. EER modeling does not support recursive subclassing.

- A. True
  - B. False
- 

98. In a disjoint, partial specialization, an entity may belong to zero or one subclass.

- A. True
  - B. False
- 

99. Categories are represented using the same notation as shared subclasses in EER diagrams.

- A. True
  - B. False
-

100. The EER model is an extension of the basic ER model.

- A. True
  - B. False
- 

## **Answer Key**

### **Multiple Choice (1–70)**

- 1. b
- 2. c
- 3. b
- 4. c
- 5. c
- 6. c
- 7. b
- 8. c
- 9. c
- 10. c
- 11. b
- 12. c
- 13. b
- 14. b
- 15. c
- 16. c
- 17. c
- 18. c
- 19. d
- 20. b
- 21. b
- 22. c
- 23. c
- 24. c
- 25. b
- 26. c
- 27. b
- 28. b
- 29. c
- 30. c
- 31. c

32. b

33. b

34. b

35. c

36. a

37. b

38. c

39. b

40. b

41. d

42. a

43. c

44. c

45. b

46. b

47. b

48. b

49. a

50. c

51. b

52. b

53. c

54. d

55. c

56. b

57. c

58. c

59. d

60. b

61. b

62. b

63. c

64. c

65. b

66. b

67. c

68. b

69. a

70. c

## **True/False (71–100)**

- 71. B
  - 72. A
  - 73. B
  - 74. A
  - 75. B
  - 76. B
  - 77. A
  - 78. B
  - 79. A
  - 80. B
  - 81. B
  - 82. B
  - 83. A
  - 84. B
  - 85. B
  - 86. A
  - 87. A
  - 88. A
  - 89. A
  - 90. A
  - 91. B
  - 92. A
  - 93. B
  - 94. B
  - 95. A
  - 96. A
  - 97. B
  - 98. A
  - 99. B
  - 100. A
-